rtab-map Localization and Navigation

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1. Course Content

1. Run the example program, and navigate through rtab-map localization mode, replacing the default amcl localization method of nav2

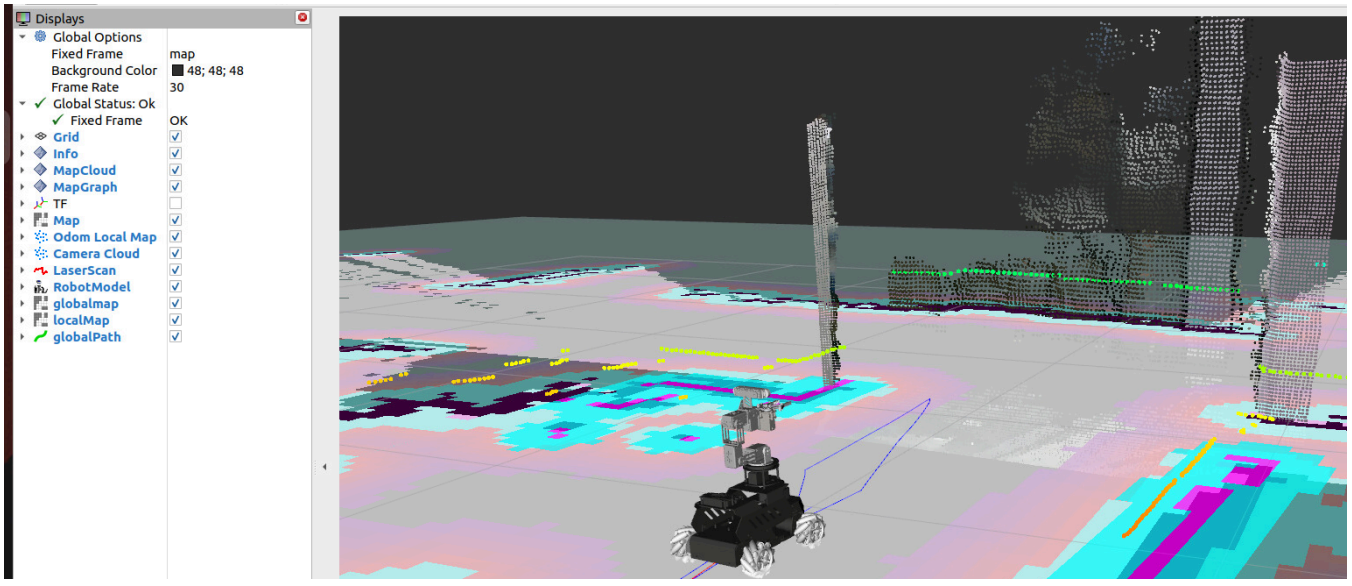
2. Principle Introduction

- Navigation 2's default localization algorithm is AMCL (Adaptive Monte Carlo Localization). The default AMCL localization method is replaced with the rtab-map localization method, while the other navigation functions remain unchanged.

3.Preparation

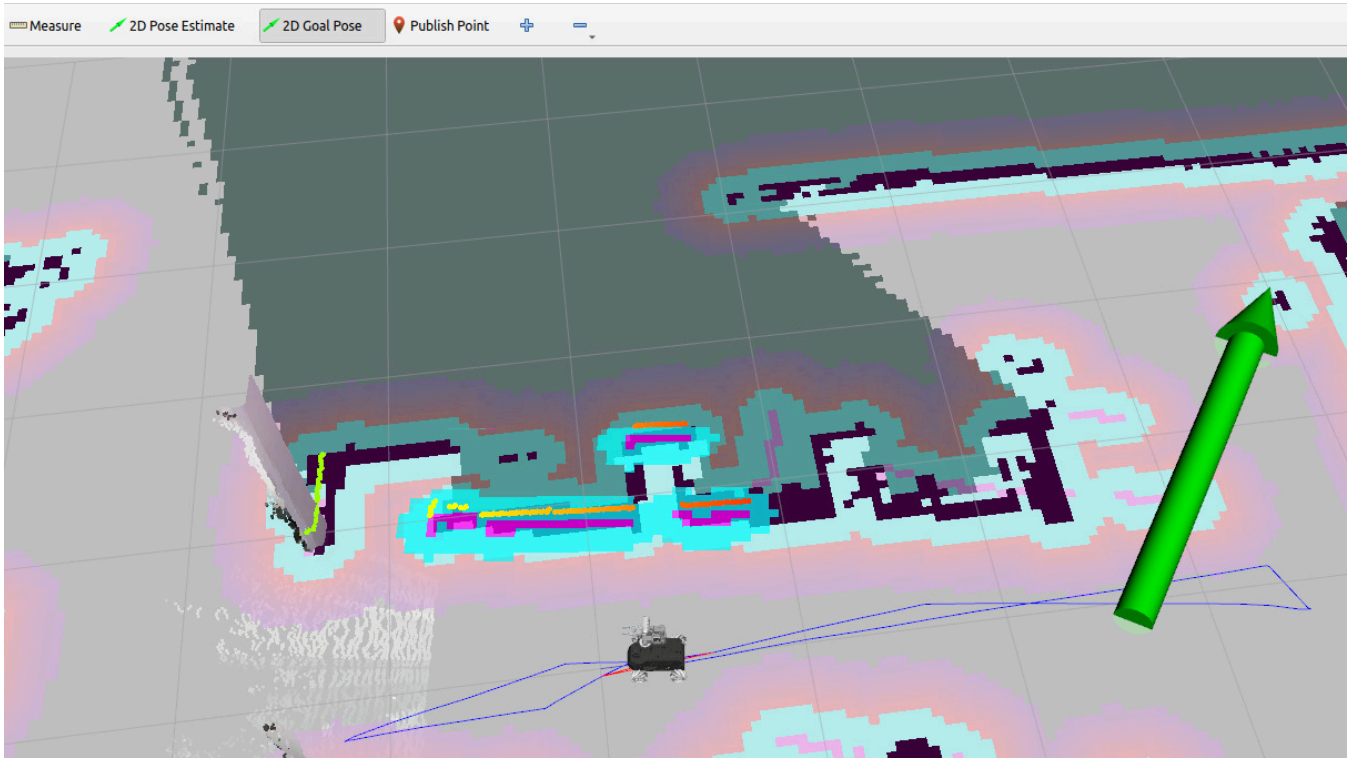
- You need to first complete the construction of a `.db`-format map; refer to the `rtab-map Algorithm Mapping` section.
- Start the program

```
ros2 launch yahboomcar_bringup nav2.launch.py localization_mode:=rtabmap
```



4. Case Demonstration

- Navigation likewise uses `2D Goal Pose` to specify the target pose, and then the car automatically plans a route to reach the destination.

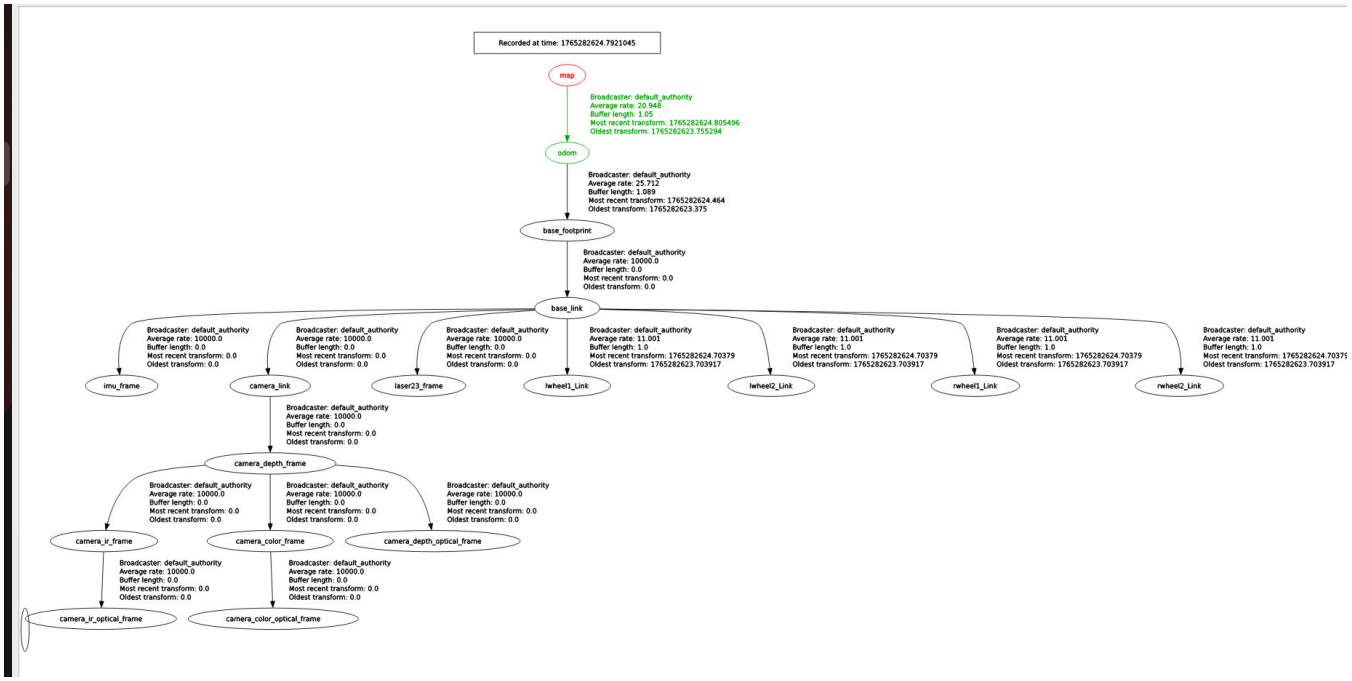


5. Source Code Analysis

5.1 View the TF Tree

```
ros2 run rqt_tf_tree rqt_tf_tree
```

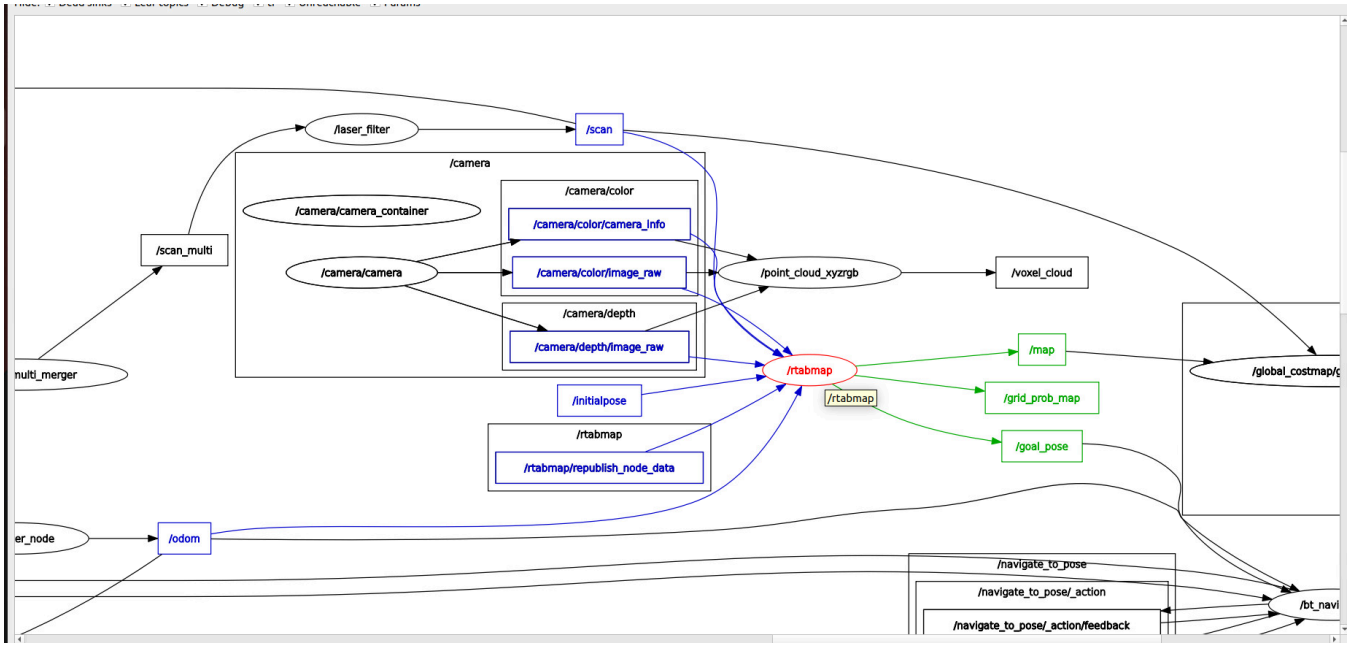
- Here, the TF transform from the map coordinate frame to the odom coordinate frame in the TF tree is provided by the rtabmap node.



5.2 View the Node Communication Graph

```
ros2 run rqt_graph rqt_graph
```

- In the figure below, when the `rtabmap` node is in localization mode, it receives the LiDAR `/scan` and odometry `/odom` topics, publishes TF transforms, and updates the `/map` map.



5.3 launch Startup File

Source code path:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/launch/nav2.launch.py
```

- The key program fragment is as follows, starting the navigation2 navigation stack node and the rtabmap localization node

```
    #rtabmap positioning navigation
    rtabmap_localization_nav = GroupAction(
        condition=IfCondition(PythonExpression(["'", localization_mode, "' == 'rtabmap' and
'", slam, "' == 'False' "])),
        actions=[
            IncludeLaunchDescription(
                PythonLaunchDescriptionSource(nav2_launch_mod),
                launch_arguments={
                                "map": map_file,
                                "params_file": params_file,
                                "use_sim_time": "False",
                                }.items()),
            IncludeLaunchDescription(
                PythonLaunchDescriptionSource(rtabmap_localization_launch_file),
                launch_arguments={
                    "mode": "localization",
                }.items(),
            )
        ]
    )
```

- Path of the rtab_map localization node startup file:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/launch/rtab_map.launch.py
```

- The key startup fragment is as follows, switching to localization mode through the localization startup parameter

```
    delay_start_rtabmap_localization= TimerAction(
        condition=IfCondition(PythonExpression(["'", mode, "' == 'localization'" ])),
        period=5.0,
        actions=[
            IncludeLaunchDescription(
                    PythonLaunchDescriptionSource(rtabmap_launch),
                    launch_arguments={
                        "rgb_topic":                    rgb_topic,
                        "depth_topic":                  depth_topic,
                        "camera_info_topic":            camera_info_topic,
                        "frame_id":                     'base_link',
                        "odom_topic":                   '/odom',
```

```
                    "use_sim_time":                 'false',
                    "rviz":                         'true',
                    "rviz_cfg":                      nav_cfg,
                    "database_path":                 database_path,
                    "rtabmap_viz":                  'false',
                    "approx_sync":                  'true',
                    "approx_sync_max_interval":     '0.01',
                    "qos":                          '0',
                    "visual_odometry":              'false',
                    "icp_odometry":                 'false',
                    "subscribe_scan":               'true',
                    "sync_queue_size":              '50',
                    "topic_queue_size":             '50',
                    "localization":                 'true',#
                    "--Mem/IncrementalMemory":      'false',
                    "log_level":                    'info'
                }.items(),
            )
    ]
)
```